

CLAUDIUS I

Wiring Diagram — Arduino Nano V3.0 (ATmega328PB, USB-C)
no RTC module in this build

Plainstack technologies · Chaos into systems.

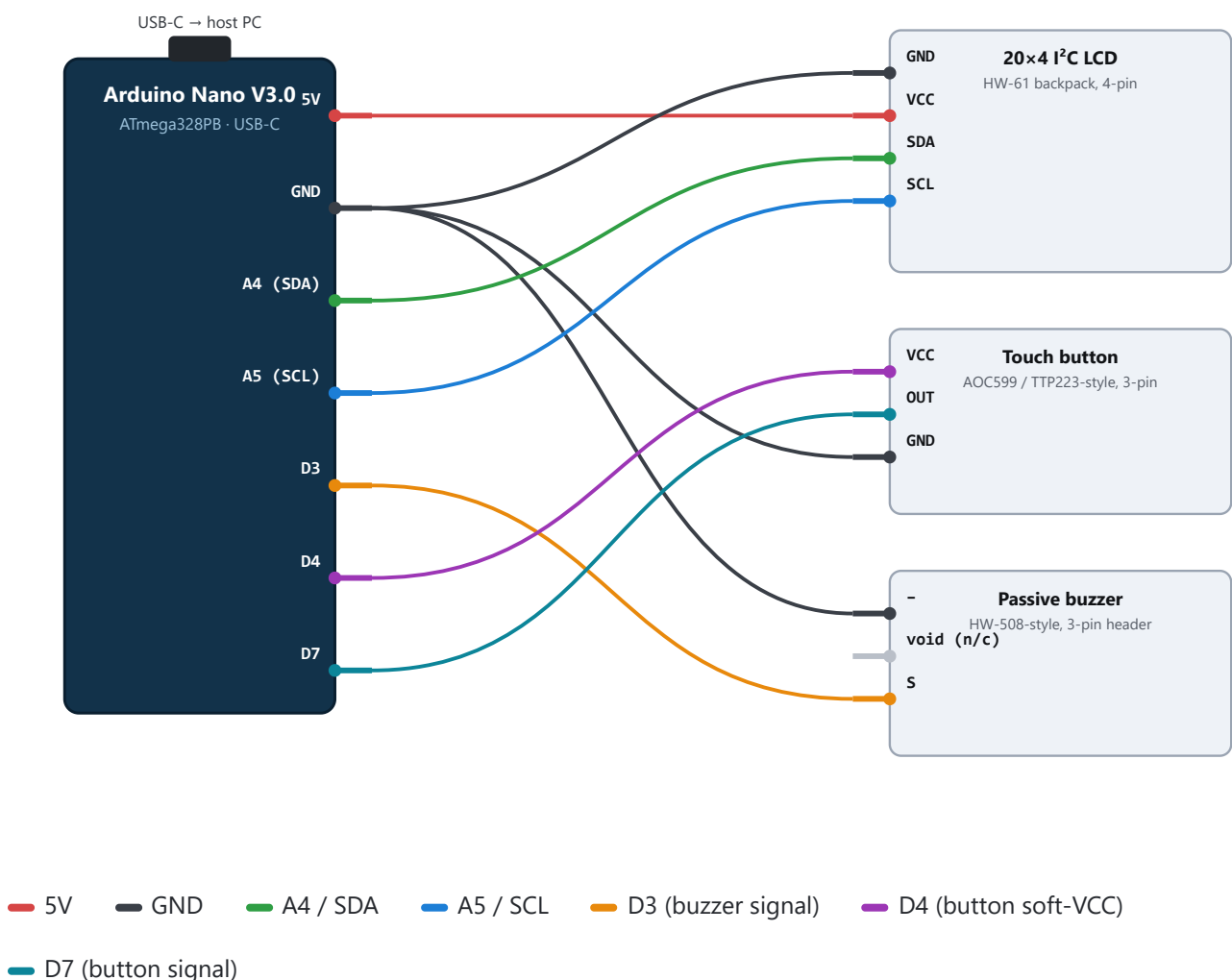
2026

1. Parts list

- Arduino Nano V3.0, ATmega328PB, USB-C connector
- 20×4 character LCD on an HW-61 I²C backpack (4-pin: GND/VCC/SDA/SCL) — address auto-tried 0x27, falls back to 0x3F
- 3-pin capacitive touch button module (AOC599 / TTP223-style: VCC/OUT/GND)
- 3-pin passive piezo buzzer module (HW-508-style header: – / void / S)

No DS3231 RTC in this build. The firmware already handles that gracefully — `rtc.begin()` just returns `false` if nothing answers at I²C address 0x68, no hang, no extra wiring needed. The boot screen shows “Welcome! / Run setup to start” until the host daemon connects and starts sending live time.

2. Diagram



3. Wiring table

From (Nano pin)	To	Notes
5V	VCC	20×4 I ² C LCD
GND	GND	20×4 I ² C LCD, Touch button, Passive buzzer
A4 (SDA)	SDA	20×4 I ² C LCD
A5 (SCL)	SCL	20×4 I ² C LCD
D3	–	Passive buzzer
D4	VCC	Touch button
D7	OUT	Touch button
D7	S	Passive buzzer

5V	LCD VCC	
GND	LCD GND, button GND, buzzer "—"	Common ground — all three share this pin
A4 (SDA)	LCD SDA	I ² C bus
A5 (SCL)	LCD SCL	I ² C bus
D3	Buzzer "S" (signal)	BUZZER_PIN , driven via tone()
—	Buzzer middle pin ("void")	Not connected — leave floating
D4	Button VCC	SWITCH_VCC_PIN — driven HIGH in software as a soft 5V rail (~5 mA draw, well under D4's 20 mA source limit)
D7	Button OUT	SWITCH_PIN , configured INPUT_PULLUP . D2 is deliberately avoided — conflicts with some Nano clones' onboard RGB LED
USB-C	Host PC	Power + serial data, 9600 baud

4. Before powering on

- Check for solder bridges, especially near **5V** and **GND** pads and around the USB-C connector — a short there can destabilize the whole USB bus on connect (symptoms seen on a previous faulty unit: USB enumeration failures, and even other unrelated USB/Bluetooth devices on the same host dropping out).
- If you have a multimeter: check continuity between 5V and GND before the first power-up — should read open / very high resistance, never near-zero.
- Give every wire a gentle tug once soldered — a joint that moves is a joint that isn't done yet.
- Flash `hex/claude_status.hex` with avrdude before final assembly to confirm the board itself is good, before committing to the enclosure.